Aletheion Workflow Automation Research Plan: First 25 Operational Workflows

Overview

This report describes an initial set of 25 GitHub-oriented workflow automations required to operate Aletheion as a real, full-stack GOD-city architecture rooted in Phoenix, Arizona.
It anchors each workflow in the existing five-layer Economic & Resource Management (ERM) architecture (Edge Sensing, State Modeling, Blockchain Trust, Optimization Engine, Citizen Interface) and in Phoenix-specific water, heat, and governance constraints.[1]
The 25 workflows are designed as a first slice through hundreds of eventual automations, chosen to cover water, thermal, energy, materials, mobility, biosignal consent, cross-species stewardship, urban hygiene, and governance.

Architectural anchors for workflows

Aletheion's ERM blueprint already defines key technical and ethical constraints: approved languages (Rust, C++, ALN, Lua, JavaScript, Kotlin), Digital Twin Exclusion Protocol, Googolswarm-based blockchain trust, and Phoenix-calibrated modules for water, heat, energy, and materials.[1]
Each workflow below is framed as a multi-layer pipeline that touches at least three of the five layers, ensuring it is not a cosmetic dashboard but a control loop tied to real infrastructure and legal constraints.[1]
Workflow definitions assume integration with centralized compliance utilities (blacklist scanning, neurorights checks) and with Indigenous land rights protocols embedded into the ERM water and land-use modules.[1]

Selection principles for the first 25 workflows

The first 25 workflows are selected using four principles: 1) start from existential constraints (water, heat, Indigenous rights), 2) cover all ERM layers, 3) ensure each workflow can map to concrete code directories and CI pipelines, and 4) maximize extensibility toward hundreds of follow-on automations.
Several workflows directly extend the existing Water Allocation and Resilience Monitoring module, Integrated Water and Thermal Flow Monitoring, and Environmental Climate Integration specification that are already defined for Phoenix.[1]
Other workflows operationalize higher-level superpowers such as Synthexis, LexEthos, CryptoSomatic Shield, Thermaphora, and Somaplex by defining how they run as repeatable jobs instead of remaining aspirational concepts.[1]

Foundational ERM and compliance workflows

1. ERM state-model sync orchestrator

Purpose: Maintain continuous, loss-aware synchronization between Layer 1 edge sensing (IoT, SCADA) and Layer 2 state model for water, thermal, energy, and materials, ensuring the operational mirror never drifts from Phoenix's physical infrastructure.[1]
Research elements: survey existing Phoenix SCADA/telemetry schemas for water plants, cool pavement sensors, and grid meters; characterize acceptable latency and data loss windows; define reconciliation strategies for conflicting measurements; and model failure modes under monsoon floods and haboobs.
GitHub focus: central orchestration service (e.g., `aletheion/erm-sync-orchestrator`) with language-specific collectors and integration tests that replay real Phoenix weather and load traces.

2. Centralized compliance preflight pipeline

Purpose: Run the Digital Twin Exclusion Protocol, blacklist scans, and neurorights compliance checks as a mandatory pre-deployment workflow for every ERM and superpower module, rather than scattering checks per-file.[1]
Research elements: formalize forbidden terminology and primitives, extend them with smart-city specific anti-patterns, codify FEAR/PAIN/SANITY envelope rules for data handling, and define FPIC validation hooks for any module touching Indigenous territories or biosignal data.[1]
GitHub focus: shared library and CI workflow definitions (e.g., `.github/workflows/compliance-preflight.yml`) invoked by all Aletheion repositories.

3. Neurorights envelope guardrail runner

Purpose: Continuously audit all biosignal, behavioral, and augmentation-related data flows for violations of CryptoSomatic Shield constraints, ensuring FEAR, PAIN, and SANITY envelopes are never breached by analytics, optimization, or monetization.[1]
Research elements: map all biosignal sources (wearables, BCIs, clinic devices), define safe feature spaces, enumerate disallowed inferences, and specify audit log schemas that are legally meaningful and computationally cheap to verify.
GitHub focus: dedicated guardrail service and scheduled jobs (e.g., `aletheion/compliance/neurorights-guard`) with integration into blockchain logging.

4. Data provenance and blockchain append workflow

Purpose: Write all significant resource transactions (water allocations, energy trades, material shipments, micro-treaties) into the Googolswarm-compatible blockchain trust layer with deterministic ordering and multi-sig attestations.[1]
Research elements: model transaction types for water, energy, and materials; define minimal, privacy-preserving payloads; specify consensus and finality requirements for city operations; and align with ALN/KYC/DID expectations for citizen and device identities.
GitHub focus: smart-contract libraries and CI pipelines for schema

validation, gas-cost simulations, and backward-compatible contract upgrades.

5. Citywide metrics and progress tracker update

Purpose: Aggregate ERM metrics (water reuse fraction, heat-vulnerability reductions, waste diversion, energy renewables share) into a versioned, auditable progress dashboard aligned with Phoenix's 2050 goals.[1]
Research elements: codify target metrics (e.g., water reuse percentages, 99% waste diversion), define aggregation windows, ensure inclusion of Indigenous and equity indicators, and correlate metrics with climatic and economic conditions.
GitHub focus: metrics aggregator service, data schemas, and automated report generation workflows publishing to both machine-readable APIs and human dashboards.

Water and thermal management workflows

6. AWP plant telemetry ingestion workflow

Purpose: Ingest, normalize, and validate high-frequency telemetry from Phoenix's Advanced Purified Water (AWP) facilities (Cave Creek, North Gateway, 91st Avenue) into the water state model with explicit capacity and quality constraints.[1]
Research elements: collect AWP plant design documents and APIs, define maximum and operational capacities (MGD), capture outage modes, and enumerate regulatory quality checks for purified potable reuse.
GitHub focus: ingestion connectors (Rust/C++), schema definitions, replay harnesses using historical Phoenix AWP-like data, and CI jobs that validate parsing against synthetic failure scenarios.

7. Groundwater recharge and aquifer monitoring workflow

Purpose: Track managed aquifer recharge projects and groundwater storage, ingesting recharge rates and storage volumes to support dryland water security and long-term portfolio planning.[1]
Research elements: analyze recharge basin designs around Phoenix,

study Colorado River flood-spreading strategies in arid cities, and derive safe operating thresholds for aquifer drawdown and recharge.
GitHub focus: recharge-site models, ingestion pipelines, and calibration tests that compare modeled storage against measured well data over multi-year horizons.

8. Water allocation optimization run (AWP-centric)

Purpose: Periodically run the AWPWaterAllocationOptimizer to allocate advanced purified water across demand zones (cool corridors, vulnerable neighborhoods, industrial users) while meeting reuse, groundwater, and Colorado River portfolio targets.[1]
Research elements: validate optimization objectives and constraints against Phoenix policy, quantify heat-vulnerability weights by neighborhood, and simulate allocations under extreme drought and outage scenarios.
GitHub focus: optimization engine modules, scenario libraries, and CI workflows that execute nightly and on-demand optimization runs with archived inputs/outputs.

9. Drought alert and demand-curtailment orchestration

Purpose: Detect approaching drought or infrastructure stress (reservoir thresholds, AWP outages) and orchestrate demand-curtailment actions across residential, commercial, and agricultural users in a legally compliant and equitably distributed manner.[1]
Research elements: integrate drought indices, reservoir and portfolio data, regulatory triggers, and elasticity of demand across sectors; co-design enforcement mechanisms that respect LexEthos and Indigenous rights.
GitHub focus: alerting logic, multi-channel notification workflows, policy-rule configuration, and simulation harnesses to test false-positive and false-negative impacts.

10. Integrated water–thermal co-optimization workflow

Purpose: Jointly optimize water allocations and thermal mitigation assets (cool pavements, shade trees, evaporative systems) to reduce urban heat while preserving scarce water, especially during extreme heat events.[1]
Research elements: quantify cooling efficacy of water-intensive

versus material-based interventions in Phoenix (e.g., irrigation vs cool pavements), model spatial correlations between water use and heat relief, and validate strategies with sensor data.
GitHub focus: co-optimization models, GIS integration modules, and scheduled analysis workflows that recommend asset deployment or reconfiguration.

Energy and materials subsystem workflows

11. Solar and grid telemetry ingestion workflow

Purpose: Ingest and reconcile real-time data from solar installations and grid meters, supporting energy balancing, demand forecasting, and resilience planning for Phoenix's high-solar context.[1]
Research elements: catalog solar assets (utility-scale, rooftop), characterize typical daily and seasonal generation profiles, and understand grid operator telemetry formats and latency characteristics.
GitHub focus: energy ingestion services, normalization code, and multi-day replay tests under heatwave and monsoon conditions.

12. Microgrid balancing and peer-to-peer settlement workflow

Purpose: Run periodic balancing and settlement across neighborhood microgrids and prosumer clusters using ALN smart contracts, ensuring that peer-to-peer energy trades respect grid stability and regulatory caps.[1]
Research elements: map regulatory limits on P2P energy trading, quantify microgrid constraints in Phoenix neighborhoods, and design pricing/settlement mechanisms that avoid burdening low-income and heat-vulnerable residents.
GitHub focus: smart-contract repositories, balancing engines, and CI workflows that simulate stress scenarios (faults, sudden demand spikes).

13. Embodied carbon and material provenance tracker

Purpose: Track the embodied carbon and provenance of construction and infrastructure materials (including cool pavements) from procurement through installation and end-of-life, supporting circular-economy and conflict-mineral policies.[1]
Research elements: catalog material supply chains for major Phoenix projects, align with lifecycle assessment standards, and identify realistic reuse/recycling pathways for asphalt, concrete, and metals.
GitHub focus: material-ledger modules, RFID/NFC integration code, and batch jobs that reconcile deliveries, installations, and demolitions.

14. Circular construction and demolition recovery orchestration

Purpose: Automate material recovery planning for construction and demolition (C&D) projects, suggesting reuse, refurbishment, or recycling routes that align with Aletheion's 99% waste-diversion targets.[1]
Research elements: study Phoenix C&D waste streams, identify local recycling/reuse capacities, and model logistics and contamination constraints that affect circularity.
GitHub focus: orchestration workflows that generate project-specific recovery plans, track compliance, and feed outcomes back into the materials state model.

Environmental climate and heat governance workflows

15. Urban heatwave early warning and cooling deployment

Purpose: Detect impending extreme heat events and orchestrate deployment of cooling assets (cool corridors, misting systems, cooling centers, schedule shifts) prioritized by Thermaphora heat-budget and vulnerability profiles.[1]
Research elements: analyze Phoenix heatwave statistics, develop HeatBudget profiles by demographic and neighborhood, and inventory cooling assets and their ramp-up characteristics.

GitHub focus: heat forecasting integration, asset deployment planners, and scenario-based CI tests that emulate historical heatwave events.

16. Cool pavement and shade asset lifecycle management

Purpose: Manage the full lifecycle of cool pavements and shade trees/canopies, from siting and installation through performance monitoring and maintenance, tied to actual surface-temperature reductions observed in Phoenix.[1]
Research elements: gather empirical data on cool pavement performance, tree survival rates, and maintenance burdens under desert conditions; encode degradation models and intervention thresholds.
GitHub focus: asset registries, sensor-association logic, and scheduled evaluation workflows that trigger maintenance tickets.

17. Dust-storm (haboob) detection and mode-shift workflow

Purpose: Detect haboobs and high-particulate events using regional sensors and forecasts, then orchestrate rapid mode shifts for transit, outdoor work, HVAC, and filtration in sensitive facilities.[1]
Research elements: compile Phoenix dust-storm history, understand transportation and health impacts, and define safe operating envelopes for equipment and human exposure.
GitHub focus: detection integrations, rule-based and learned response engines, and regression tests replaying known historic events.

Mobility, Somaplex, and logistics workflows

18. Mobility demand sensing and multimodal routing orchestration

Purpose: Integrate GTFS transit data, traffic sensors, micromobility telemetry, and pedestrian counts into adaptive routing recommendations that balance speed, safety, emissions, and heat exposure.[1]
Research elements: obtain Phoenix transit and traffic data, study

modal splits and peak patterns, and quantify trade-offs between speed and health (e.g., shade vs direct sun walking).
GitHub focus: routing engines, interface APIs, and batch simulations that compare routing strategies across typical and extreme days.

19. SomaticRouteEngine update workflow

Purpose: Periodically recompute SomaticRouteEngine parameters and route suggestions based on updated joint-load, fall-risk, and heat-exposure data, especially for elders, disabled residents, and heat-vulnerable workers.[1]
Research elements: map fall incidents, joint-pain reports, and microclimate variations across Phoenix; develop ergonomic and clinical models that translate into route penalties and preferences.
GitHub focus: Somaplex models, data pipelines, and scheduled jobs that recompute per-person and per-cohort route preferences.

20. Last-mile logistics and freight separation orchestrator

Purpose: Separate heavy freight from residential and pedestrian flows via time-windowing, routing, and curb-space allocation, minimizing noise, emissions, and conflict in vulnerable neighborhoods.[1]
Research elements: study freight patterns in Phoenix, warehouse locations, delivery windows, and existing noise/air-quality regulations; quantify co-benefits of time-shifting deliveries.
GitHub focus: freight routing optimizers, curb management modules, and workflows that generate schedules and enforcement feeds.

Neurobiome, Synthexis, and LexEthos workflows

21. Microbiome-corridor monitoring and cleaning-protocol adjustment

Purpose: Monitor Neurobiome corridors (gut, skin, air, soil) and automatically adjust cleaning agents and surface materials to protect beneficial microbiota while suppressing pathogens in homes, clinics, and transit.[1]

Research elements: inventory surface materials and cleaning protocols across city facilities, collaborate with microbiome scientists to define protective vs harmful profiles, and model exposure loops.
GitHub focus: microbiome score calculators, surface/agent registries, and batch workflows that propose cleaning schedule and agent changes.

22. Synthexis LightNoisePesticidePlanner nightly run

Purpose: Execute the LightNoisePesticidePlanner to compute operating envelopes for lighting, noise, and pesticide use that satisfy BioticTreaties while maintaining human comfort and safety.[1]
Research elements: collect species-activity forecasts for bats, pollinators, and desert biota around Phoenix; gather human comfort and safety thresholds; and model corridor continuity and conflict zones.
GitHub focus: Synthexis planner modules, species-activity data ingestors, and nightly jobs that publish recommended envelopes and violation reports.

23. LexEthos micro-treaty compilation and deployment workflow

Purpose: Compile neighborhood- and workplace-level rights grammars (e.g., noise limits, shade access, augmentation boundaries) into executable micro-treaties and deploy them to relevant devices and services.[1]
Research elements: co-design rights grammars with communities, map them to enforceable device constraints, and study dispute patterns to refine grammar expressiveness.
GitHub focus: LexEthos compilers, treaty-distribution services, and regression tests that ensure device behavior conforms to active micro-treaties.

Citizen interface, consent, and grievance workflows

24. Consent-state synchronization and audit-trail workflow

Purpose: Synchronize fine-grained consent states for biosignals, location, augmentation, and participation in experiments across all citizen-facing apps and devices, maintaining an immutable audit trail in the trust layer.[1]
Research elements: specify consent-state machines per data type, align with legal and neurorights requirements, and test revocation, expiry, and emergency-override behaviors.
GitHub focus: consent APIs, DID integration, and scheduled reconciliation jobs that detect and rectify drift between declared and effective consent.

25. Incident, grievance, and cooling-resolution orchestrator

Purpose: Provide a unified intake, triage, and resolution workflow for incidents and grievances related to water, heat, mobility, augmentation, and rights violations, with a built-in “cooling” protocol that favors de-escalation and restorative actions.[1]
Research elements: analyze historical complaint and incident patterns in Phoenix, work with communities to define desired remedies, and encode escalation ladders that avoid policing-first responses.
GitHub focus: grievance intake services, LexEthos-compatible resolution engines, and periodic analytics jobs that surface systemic issues for policy change.

Extensibility toward hundreds of workflows

The 25 workflows above create a backbone for hundreds of additional automations by establishing patterns for ingestion, optimization, treaty enforcement, consent management, and cross-species stewardship across all ERM layers.[1]

Future workflows can be derived by combining and specializing these patterns for education, culture, manufacturing, finance, and emergency response while remaining grounded in Phoenix's infrastructure and Aletheion's exclusion and neurorights protocols.[1]
Every new workflow should be added as a dedicated GitHub path with clear ownership, CI workflows, and explicit links back to the ERM architecture and the city's legal and ecological operating envelopes.[1]

References

1. here-is-the-first-version-of-i-u2wLYLdjRuiC_oi5Jgzpmw.md - img srchttpsr2cdn.perplexity.aipplx-full-logo-primary-dark402x.png styleheight64pxmargin-right32px